DRIVEABILITY—CNG BI-FUEL—NO START, LONG CRANK, HESITATION OR REDUCED ENGINE POWER—WHILE OPERATING IN BI-FUEL MODE ONLY—VEHICLES EQUIPPED WITH 5.4L CNG BI-FUEL ENGINE ONLY	Article No. 01-24-4

FORD: 2000-2001 F-150

This article is being republished in its entirety to update the Calibration Information Chart.

ISSUE

While operating in the alternate fuel mode only, some vehicles may exhibit any of the following conditions:

- No start
- Crank time greater than 10 seconds while attempting to start
- Reduced engine power with power steering application at idle or moving less than 5 mph
- Reduced engine power on decel after cresting a steep hill at 10 mph or less

This may be caused by the calibration of the Powertrain Control Module (PCM).

ACTION

Verify the condition. If normal diagnostics cannot pinpoint the condition, reprogram the PCM to the latest level calibration. If the new calibration does not correct the condition:

- Replace the compuvalve and the MAP line tube assembly on 2000 CNG vehicles
- Replace the compuvalve only on the 2001 CNG bi-fuel vehicles.

NOTE

REPAIR ACTION APPLIES ONLY TO CNG OPERATION. SHOULD THESE CONDITIONS OCCUR WHILE OPERATING IN THE GASOLINE MODE, FOLLOW NORMAL WORKSHOP MANUAL DIAGNOSTIC PROCEDURES.

SERVICE PROCEDURE

1. Reprogram the PCM. Use the Ford calibration installation procedure as specified in the WDS screen commands.
2. Verify that the condition has been corrected.
3. If the condition has been corrected, the repair action is complete. If the condition has not been corrected, continue with Step 4.
4. Monitor:
   a. Use NGS or WDS to monitor the ADPTCTL (Fuel Adaptive Control) PID from the AFCM.
   b. Turn off all accessories. e.g. heated rear screen, radio, lights, air conditioning, and passenger compartment fan (if equipped).
   c. With the foot brake fully applied, idle the vehicle in Drive and apply a full power steering load. Release the steering wheel and allow engine idle to stabilize.
   d. With the foot brake fully applied, idle the vehicle in Drive and apply a full power steering load for the second time. Note the value of the ADPTCTL PID when the steering is at full load.
   e. Release the steering wheel and allow engine idle to stabilize. Note the value of the ADPTCTL PID.
   f. If the ADPTCTL PID did not change following the second full power steering load application and vehicle does stall, follow normal Workshop Manual diagnostic procedures to resolve the condition. If the ADPTCTL PID changes more than ±0.15 following the second full power steering load application, continue with Step 5.
5. Compuvalve:
   a. For the 2000 CNG bi-fuel vehicle, replace the compuvalve using the Ford removal and installation - Compuvalve Natural Gas procedure as specified in the 2000 F150 Workshop Information section 303-04E. Also, replace the MAP line tube assembly as specified in Step 6.

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b. For the 2001 CNG bi-fuel vehicle, replace the compuvalve using the Ford removal and installation - Compuvalve Natural Gas procedure as specified in the 2001 Truck Workshop Information section 303-04D.

6. Map Line:

a. Locate MAP line tube assembly. One end is connected to the MAP barb of the compuvalve, the other end is a TEE going into the canister purge delivery line (to the back side of the intake manifold inlet) and to the rear of the throttle body (Figure 1).

b. Remove the MAP line tap assembly. Remove the hoses off each end of the TEE and off the barb of the compuvalve.

c. Install new MAP line tap assembly. Reconnect TEE and to the barb of the compuvalve. Route the MAP line so that it does not come in contact with the bulkhead behind the engine (Figure 1).

PCM CALIBRATION INFORMATION

Application	Old Part Number (-12A650-)	Tear Tag	New Part Number (-12A650-)	Old Calibration	New Calibration	NGS/WDS Qualifier
2000 Bi-Fuel F-150 3.55 Axle Ratio	YL3F-ARA	PMZ1	YU7A-AVB	0F51410B00	0F51410B00	WDS Release B16.2 or Later
2000 Bi-Fuel F-150 3.73 Axle Ratio	YL3F-ASA	GZK1	YU7A-AXB	0F51420B00	0F51420B00	WDS Release B16.2 or Later
2001 Bi-Fuel F-150 3.55 Axle Ratio	1L3A-BAB	KVE2	1L3A-BAD	1F51410B10	1F51410B10	WDS Release B16.2 or Later
2001 Bi-Fuel F-150 3.55 Axle Ratio	1L3A-BAC	KVE2	1L3A-BAD	1F51410B11	1F51410B11	WDS Release B16.2 or Later
2001 Bi-Fuel F-150 3.73 Axle Ratio	1L3A-BBB	PQJ2	1L3A-BBD	1F51420B10	1F51420B10	WDS Release B16.2 or Later
2001 Bi-Fuel F-150 3.73 Axle Ratio	1L3A-BBC	PQJ2	1L3A-BBD	1F51420B11	1F51420B11	WDS Release B16.2 or Later

PART NUMBER	PART NAME
1L3Z-9L474-AA	Map Damper Tube
YL3Z-9F954-SARM	Compuvalve - 2000 F-150 CNG Bi-Fuel
1L3Z-9F954-BARM	Compuvalve - 2001 F-150 CNG Bi-Fuel

Obtain an Authorized Modifications Decal (STAT 5941-2 - obtainable through the PDC) and list the date, dealer number, and summary of modifications performed. Select a prominent place adjacent to the Vehicle Emission Control Information Decal suitable for installing the Authorized Modifications Decal. Clean the area, install the decal, and cover it with a clear plastic decal shield.

Ford AUTHORIZED MODIFICATIONS

THE FOLLOWING MODIFICATIONS HAVE BEEN MADE:

Reprogrammed Powertrain Control Module (PCM) Per TSB 01-24-4

THESE MODIFICATIONS HAVE BEEN APPROVED, AS APPROPRIATE, BY EPA AND CARB.

DEALER NUMBER: **DATE:**

CHANGE AUTHORITY:

FPS 8262 9/78 **FORD MOTOR COMPANY** **PRINTED IN U.S.A.**

OTHER APPLICABLE ARTICLES: NONE
SUPERSEDES: 01-16-5
WARRANTY STATUS: Eligible Under The Provisions Of Bumper To Bumper Warranty Coverage And Emissions Warranty Coverage

OPERATION	DESCRIPTION	TIME
012404A	Reprogram Powertrain Control Module (PCM)	0.7 Hr.
012404B	Reprogram Powertrain Control Module And Replace Compuvalve (Includes Time To Replace Map Line)	1.7 Hrs.

DEALER CODING

BASIC PART NO.	CONDITION CODE
RECALEM	42

OASIS CODES: 601300, 602300, 603300, 607000, 607400, 607500, 607600, 607700, 617400, 618400, 619400

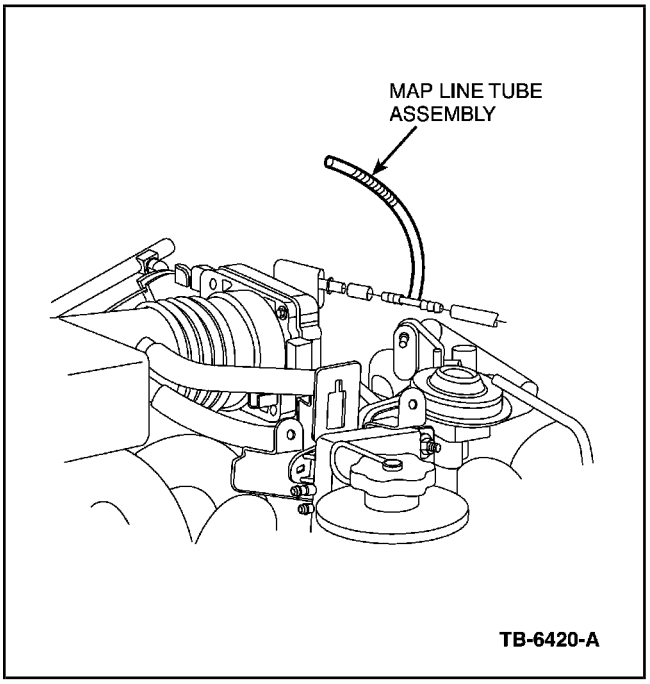


Figure 1 - Article 01-24-4